KIIT UNIVERSITY

Scheme of Evaluation

Autumn End Semester Examination-2025
5th Semester B.Tech
ENGINEERING ECONOMICS HS30101
Full Mark-50

1.	a.	Substitutes When P_y increase Q_x increases	[1×10]
	b.	P = 25 Q = 50	
	c.	$M = P_x.X + P_y.Y$ (budget equation) $\frac{P_x}{P_y}$ (slope of the budget line)	
	d.	Relation between TP and MP	
	e.	Increasing K and reducing L	
	f.	Final Amount (F) = 69211.38815	
	g.	Effective Rate = 8.16% ∵ 8% nominal interest compounded semi-annually = Effective Annual Rate of 8.16%	
	h.	B_{13} = 475518.9031	
	i.	• Equilibrium conditions for a perfectly competitive firm. • Equilibrium condition with MR and MC curve.	
	j.	Effect of rise in bank rate on your borrowing from banks	
2.	a.	(i) % ΔQ for petrol = –5% % ΔP for petrol = 20% $e = \frac{5}{20} = -0.25$ Petrol is less elastic in Metro cities. (ii) • Two important factors influencing the degree of price elasticity f petrol. • One suggestion to make the consumers demand more elastic for petrol.	[5]
	b.	(i) Sales of S_{25} ultra increases by 14% due to increase in per capita income and increase in price of Apple iphone16. (ii) • Price elasticity for S_{25} Ultra = $\frac{6}{-5} = -1.2$	[5]

		• Samsung Electronics should reduce the price to raise the TR	
3.	a.	(i) • Before tax P = 100 Qd = Qs = 400 • After tax Qs = 140 + 2P P = 106 Qd = Qs = 352 • Producer is taking a higher tax burden (ii) Distribution of tax burden with a suitable diagram.	[5]
	b.	(i) Single deposit (P) Junali has to make P = 20 00 000 (P/F 7.5% 8) P = 1121404.467 (ii) • Annual Equivalent amount (A) A = 1121404.467 (A/P 7.5% 8) A = 191454.0464 • Check F = 191454.0464 (F/A 7.5% 8) = 20 00 000	[5]
4.	a.	NPV (12%) = –100 00 00 000 + 10 00 00 000 (P/A 12% 10) +12 00 00 000 (P/A 12% 10) × (P/F 12% 10) +20 00 00 000 (P/F 12% 20) = –195937872.6 Thus, KIMS should not go ahead with the proposal.	[5]
	b.	NPV (12%) = –10 00 000 + 3 20 000 (P/A 12% 4) +100 000 (P/F 12% 5) = 28694.47649 NPV (14%) = –10 00 000 + 3 20 000 (P/A 14% 4) +100 000 (P/F 14% 5) = –15675.19612	[5]

		By interpolation IRR = 13.29% (approximate) Students might have tried at different percentages. Since IRR is greater than required rate, the company should purchase the machine.	
5.	a.	(i) • Increasing Return to Scale (IRS) • Figure for IRS in terms of Iso-quants (ii) • Constant Return to Scale (CRS) • Figure for CRS in terms of Iso-quants	[5]
	b.	(i) $TC = 100 + 10Q + Q^2$ $AC = \frac{100}{Q} + 10 + Q$ $MC = \frac{dC}{dQ} = 10 + 2Q$ At AC = MC Q = 10 AC = 30, MC = 30 It is the minimum point of AC as the value of AC and MC is same. (ii) • Prove that AVC is directly proportional to the reciprocal of AP_L. • Drawing the AVC curve with the help of AP_L curve. • Writing the relations between AVC and AP_L curves.	[5]
6.	a.	K = 0.4 Year / Annual Dep. / Book value 0 / 0 / 110000 1 / 44000 / 66000 2 / 26400 / 39600 3 / 15840 / 23760 4 / 9504 / 14256 5 / 5702.4 / 8553.6	[5]
	b.	AW (10%) = 200 00 000 + 100 00 000 (A/F 10% 20) Benefit = 20174596.25 AW (10%) = 20 00 00 000 (A/P 10% 20)	[5]

		Costs +3 00 00 000 (P/F 10% 10) × (A/P 10% 20) +50 00 000 = 29850498.06 BC ratio = $\frac{20174596.25}{29850498.06}$ = 0.6758 Since the BC ratio is less than 1 the project should not be implemented.	
7.	a.	(i) $Q_B = \frac{200\,000}{10} = 20\,000$ units Q_B with 20% increase in SP $= \frac{200\,000}{20} = 10\,000$ units P/V ratio = $\frac{1}{5}$ or 0.2 Breakeven sales value = $\frac{200\,000}{0.2} = 10\,00\,000$ (ii) Graphically showing the MOS when actual sales is 25000 units.	[5]
	b.	Explaining Demand-Pull and Cost-Push inflation with suitable diagrams.	[5]

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